

(1001CJA102120004)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

09-02-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	A	D	A,D	A,B,D	A,B,D	A,C,D	B,C	B,D
	Q.	11	12								
	A.	A,B,C	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.00	5.00	1.33 to 1.34	1.56 to 1.58	6.00	2.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	D	A	B,C,D	B,C,D	A,B,C,D	B,D	B,C	A,B,C,D
	Q.	11	12								
	A.	A,D	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	533.80	6.00	4.00	3.00	6.00	2.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	D	A,B	A,D	A	B,D	A,C	A,B,C,D
	Q.	11	12								
	A.	A,B,C,D	A,B,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.00	9.48 or 9.49	0.25	36.00	0.25	2.50				

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B	B,D	A,D	A,C,D	B,C	A,B	A,B,C,D	A,C,D	B	C
	Q.	11	12								
	A.	B	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.00	2.00	2.00	9.00	10.41 to 10.43	4.00				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,C	B,D	A,B,C,D	A,B,D	A,B	A,C	A,C,D	C	C
	Q.	11	12								
	A.	A	D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	3.42	2.50	3.00	4.00	5.00	6.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	B,C,D	A,C,D	A,B,C	C	B,C,D	B,D	B,D	D	B
	Q.	11	12								
	A.	D	B								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.00	0.00	1.80	8.75	3.00	0.00				

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I)**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (D)**

$$\text{Sol. } m = \frac{f}{f+u}$$

$$\Rightarrow m = \frac{+20}{+20 + (-30)} = -2$$

$$\bar{v}_0 = 8i + 6j$$

$$\bar{v}_1 = (2)^2 8i + (-2) 6j$$

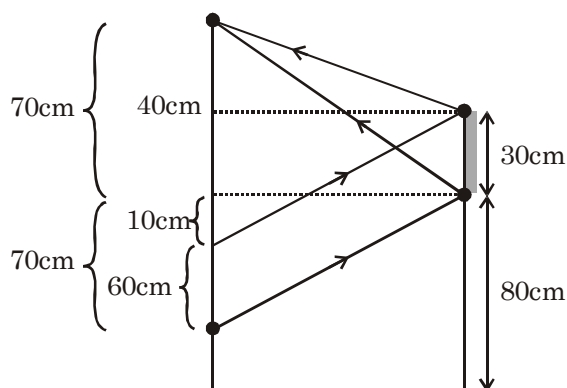
$$\bar{v}_1 = 32i - 12j$$

2. **Ans. (B)****Sol.** Present deviation

$$\delta = (i + e) - A$$

$$\delta = (53 + 37) - 60 = 30^\circ$$

so minimum deviation will be less than present deviation

3. **Ans. (A)****Sol.** The situation is clearly shown in figure.

The height of the image he sees himself is 60cm

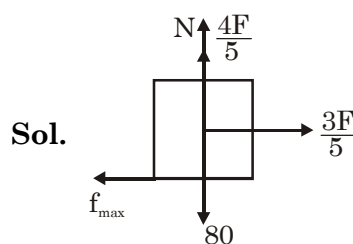
4. **Ans. (D)****Sol.** From energy conservation after spring break

$$\frac{1}{2}k\left(\frac{A}{2}\right)^2 + \frac{1}{2}m\left(\omega\frac{\sqrt{3}A}{2}\right)^2 = \frac{1}{2}kA'^2$$

$$\frac{1}{2}k\frac{A^2}{4} + \frac{1}{2}m\frac{k}{m}\frac{3A^2}{2} = \frac{1}{2}kA'^2$$

$$\frac{7A^2}{4} = A'^2$$

$$A' = \frac{\sqrt{7}A}{2}$$

5. **Ans. (A,D)**6. **Ans. (A,B,D)**7. **Ans. (A,B,D)****Sol.**

to start motion

$$\frac{3F}{5} = \mu\left(80 - \frac{4F}{5}\right)$$

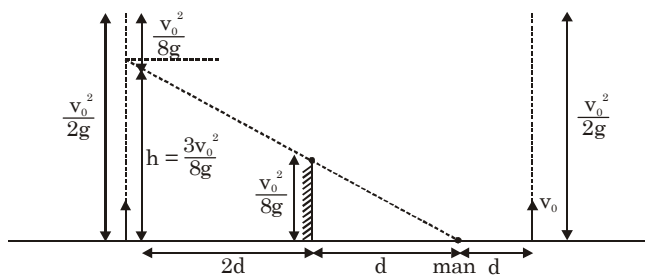
$$\Rightarrow F = 50$$

So for $F < 50$ block will be at rest & for $F > 50$ block will move

8. **Ans. (A,C,D)**9. **Ans. (B,C)****Sol.** $\delta = A(\mu - 1)$ for small prism

10. Ans. (B,D)

Sol.



from similar triangle

$$\frac{h}{3d} = \frac{v_0^2 / 8g}{d} \Rightarrow h = \frac{3v_0^2}{8g}$$

$$\text{total time of flight} = \frac{2v_0}{g}$$

time for which object will not be seen

$$= 2\sqrt{\frac{2(v_0^2/8g)}{g}} = \frac{v_0}{g}$$

time for which object will be seen

$$= \frac{2v_0}{g} - \frac{v_0}{g} = \frac{v_0}{g}$$

11. Ans. (A,B,C)

12. Ans. (A,C)

SECTION-II

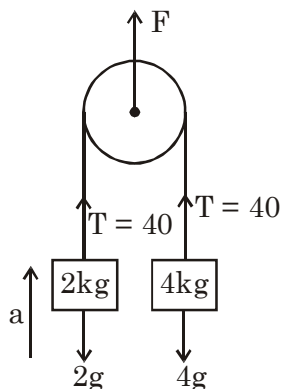
1. Ans. 3.00

Sol. $\omega = \sqrt{\frac{g}{R \cos \theta}}$

$$2 = \sqrt{\frac{10}{5 \cos \theta}}$$

$$\theta = 60^\circ = \frac{\pi}{3}$$

2. Ans. 5.00



$$a = \frac{40 - 20}{2} = 10$$

$$a_{\text{pulley}} = \frac{10 + 0}{2} = 5$$

3. Ans. 1.33 to 1.34

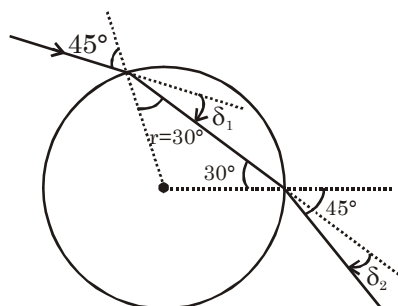
Sol. $T = \frac{2\pi}{g} \sqrt{\ell} + \left(2\pi \sqrt{\frac{\ell}{g}} \right) \frac{1}{6} = \frac{4}{3}$ seconds

4. Ans. 1.56 to 1.58

Sol. $T = 2\pi\sqrt{\frac{\mu}{k_{\text{eff}}}} = 2\pi\sqrt{\frac{20}{20}} = 2\pi$

$$\text{required time} = \frac{T}{4} = \frac{\pi}{2} \text{ seconds}$$

5. Ans. 6.00



Sol.

$$\begin{aligned}\mu_1 \sin i &= \mu_2 \sin r \text{ (Snell's law)} \\ &= (\sin 45^\circ) = \sqrt{2} \sin r\end{aligned}$$

$$\sin r = \frac{1}{2}$$

$$r = 30^\circ$$

$$\begin{aligned}\text{Deviation} &= \delta_1 + \delta_2 \\ &= 15^\circ + 15^\circ\end{aligned}$$

$$= 30^\circ = \frac{\pi}{6} \text{ radian}$$

6. Ans. 2.00

Sol. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$$\frac{\mu}{2R} - \frac{1}{\infty} = \frac{\mu - 1}{R}$$

$$\frac{\mu}{2R} = \frac{\mu - 1}{R}$$

$$\mu = 2$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (D)
2. Ans. (D)
3. Ans. (D)
4. Ans. (A)
5. Ans. (B,C,D)
6. Ans. (B,C,D)
7. Ans. (A,B,C,D)
8. Ans. (B,D)
9. Ans. (B,C)

10. Ans. (A,B,C,D)
11. Ans. (A,D)
12. Ans. (B,C,D)

SECTION-II

1. Ans. (533.80)
2. Ans. (6.00)
3. Ans. (4.00)
4. Ans. (3.00)
5. Ans. (6.00)
6. Ans. (2.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (C)
Sol. $f'''(1) = f''(2) = f'(3) = 0$
& $f'(x) = (x-1)^4 (x-2)^3 (x-3)^2 \cdot 1 + \dots$
So, $f'(4) = 3^4 \cdot 2^3 \cdot 1 = 3 \cdot 6^3$
2. Ans. (C)

Sol. $\lim_{x \rightarrow 0} \frac{x^{1/3} x^{2/3} \left(1 + \frac{1}{x^2}\right)^{1/3}}{(x^2)^{1/2} \left(a + \frac{b}{x^2} + \frac{c}{x^3}\right)^{1/2}} = 2$

$$\lim_{x \rightarrow 0} \frac{x \left(1 + \frac{1}{x^2}\right)^{1/3}}{\left(a + \frac{b}{x} + \frac{c}{x^2}\right)^{1/2}} = 2$$

$$\Rightarrow \frac{1}{a^{1/2}} = 2$$

$$\Rightarrow \frac{1}{2} = \sqrt{a}$$

$$\Rightarrow \boxed{\frac{1}{4} = a}$$

3. Ans. (A)

Sol. For $x < -1$, $\lim_{n \rightarrow \infty} \frac{(1^-)^\infty - 4}{(1^+)^\infty + 2} \quad \left(\frac{\infty}{\infty} \text{ form}\right)$

As $|x| > 1$, So, divide by $|x|^n$

$$f(x) = \lim_{n \rightarrow \infty} \frac{1 - \frac{4}{|x|^n}}{1 + \frac{2}{|x|^n}} = \frac{1-0}{1+0} = 1$$

For $x > 0$

$$\text{at } x = 0^+ \quad \frac{\sin 3x}{\sin 2x} > 1 \Rightarrow \left[\frac{\sin 3x}{\sin 2x} \right] = 1$$

for continuous at $x = 0$ & $x = 1$

$$b = 2$$

$$\text{and } -a + b = 1$$

$$\text{we get } a = 1$$

$$\therefore f\left(\frac{-1}{2}\right) = \frac{-a}{2} + b = \frac{-1}{2} + 2 = \frac{3}{2}$$

4. Ans. (D)

Sol. Let $\frac{1}{x} = y$

$$\lim_{y \rightarrow 0} \left\{ \frac{(1+y)^{1/y} - e}{y} \right\}$$

$$= \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y)} - e}{y}$$

$$= e \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y) - 1} - 1}{y}$$

$$= e \lim_{y \rightarrow 0} \frac{e^{\frac{1}{y} \ln(1+y) - 1} - 1}{\frac{1}{y} \ln(1+y) - 1} \times \frac{\ln(1+y) - y}{y^2}$$

$$= e \lim_{y \rightarrow 0} \frac{\ln(1+y) - y}{y^2} = e \lim_{y \rightarrow 0} \left(y - \frac{y^2}{2} + \dots \right) - y$$

$$= -\frac{e}{2}$$

5. Ans. (A,B)

Sol. For $x \geq 0$ $f(x) = a \cos x + be^x + cx^3$
 $f(0^+) = a + b$
 For $x < 0$ $f(x) = +a \cos x + be^{-x} + cx^3$
 $f(0^-) = a + b$
 $LHL = RHL = f(0)$
 Now for differentiability
 $f'(0^+) = f'(0^-)$
 $\Rightarrow \lim_{x \rightarrow 0^+} -a \sin x + be^x + 3cx^2$
 $= \lim_{x \rightarrow 0^-} -a \sin x - be^{-x} - 3cx^2$
 $\Rightarrow b = 0$ and a, c can take any real value

6. Ans. (A,D)

Sol. $y = \cot^{-1} \left(\frac{2 + 2\sqrt{\cos^2 x}}{2 \sin x} \right)$
 $\Rightarrow y = \cot^{-1} \left(\frac{1 + |\cos x|}{\sin x} \right)$
 For $x \in \left(0, \frac{\pi}{2} \right)$,
 $y = \cot^{-1} \left(\frac{1 + \cos x}{\sin x} \right) = \cot^{-1} \left(\cot \frac{x}{2} \right)$
 $\Rightarrow y = \frac{x}{2} \quad \therefore \frac{dy}{dx} = \frac{1}{2}$
 For, $x \in \left(\frac{\pi}{2}, \pi \right)$ $y = \cot^{-1} \left(\frac{1 - \cos x}{\sin x} \right)$
 $\Rightarrow y = \cot^{-1} \left(\tan \frac{x}{2} \right)$
 $\Rightarrow y = \cot^{-1} \left(\cot \left(\frac{\pi}{2} - \frac{x}{2} \right) \right)$
 $\Rightarrow y = \frac{\pi}{2} - \frac{x}{2}$
 $\Rightarrow \frac{dy}{dx} = -\frac{1}{2}$

7. Ans. (A)

Sol. $\frac{1}{x} = 2 \sin(\cos^{-1} x)$
 $\frac{1}{x} = 2\sqrt{1-x^2}$
 $\Rightarrow 4x^4 - 4x^2 + 1 = 0$
 $\Rightarrow x^2 = \frac{1}{2}$ (Verify also)

8. Ans. (B,D)

Sol. $f(x) = [x + \{x + \text{integer}\}] = [x + \{x\}]$
 At $x = n$; $n \in \mathbb{I}$
 $f(n^-) = \lim_{h \rightarrow 0^+} [n - h + \{n - h\}]$
 $= [n - h + 1 - h] = n$
 $f(n) = n$
 $\& f(n^+) = \lim_{h \rightarrow 0^+} [n + h + \{n + h\}] = [n + h + h] = n$
 So, $f(x)$ is continuous for $\forall x \in \text{Integer}$
 Hence, Differentiable also as $f(x) = n$
 for $x \in \left[n - \frac{1}{2}, n + \frac{1}{2} \right)$
 $f'(x) = 0 \quad \forall x \in \left[n - \frac{1}{2}, n + \frac{1}{2} \right)$

9. Ans. (A,C)

Sol. $[|\sin^{-1} x|] = 0$
 $0 \leq |\sin^{-1} x| < 1$
 $\Rightarrow -1 \leq \sin^{-1} x \leq 1$
 $\Rightarrow 0 \leq \sin^{-1} |x| \leq 1$
 $\Rightarrow 0 \leq |x| \leq \sin 1$
 $\Rightarrow -\sin 1 \leq x \leq \sin 1 \Rightarrow p = -\sin 1 \& q = \sin 1$
 $\Rightarrow p^2 + q^2 = 2\sin^2 1$
 as $2\sin^2 45^\circ < 2\sin^2 1 < 2\sin^2 60^\circ$
 $\Rightarrow 1 < 2\sin^2 1 < \frac{3}{2} \Rightarrow 1 < p^2 + q^2 < \frac{3}{2}$

10. Ans. (A,B,C,D)

Sol. $h(x) = \begin{cases} f(x) & x^2 < 1 \\ \frac{f(x) + g(x)}{2} & x^2 = 1 \\ g(x) & x^2 > 1 \end{cases}$
 $\Rightarrow h(x) = \begin{cases} x^2 + bx + 1 & x < -1 \\ (a + 2 - b)/2 & x = -1 \\ \tan \frac{(\pi(x+1))}{8} + a & -1 < x < 1 \\ \frac{1 + a + 2 + b}{2} & x = 1 \\ x^2 + bx + 1 & x > 1 \end{cases}$

Now, $f(-1^-) = f(-1) = f(-1^+)$

$$\Rightarrow 1 - b + 1 = \frac{a + 2 - b}{2} = a \Rightarrow a + b = 2$$

$$\text{and } f(1^-) = f(1) = f(1^+) = 1$$

$$\Rightarrow 1 + a = \frac{1 + a + 2 + b}{2} = 2 + b \Rightarrow a - b = 1$$

$$\text{Solving, we get } a = \frac{3}{2} \text{ \& } b = \frac{1}{2}$$

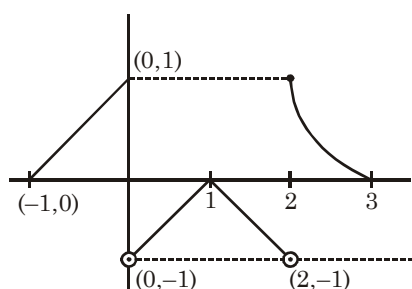
$$a^2 + b^2 = \frac{9}{4} + \frac{1}{4} = \frac{5}{2}$$

$$\& g(x) \geq \frac{-D}{4a} = \frac{-(b^2 - 4)}{4} = \frac{15}{16}$$

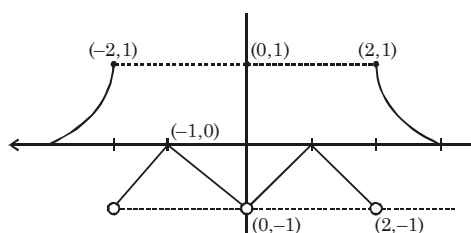
$$\text{and } f(x) \text{ is periodic with period } \frac{\pi}{8} = 8$$

11. Ans. (A,B,C,D)

Sol. Graph of $y = f(x)$



Graph of $y = f(|x|)$



12. Ans. (A,B,C)

$$\text{Sol. } L = \lim_{x \rightarrow 0^+} x^\alpha \left(\frac{1}{\log(2x + (1+x^3)^{1/2})} - \frac{1}{\log(1+2x)} \right)$$

$$= \lim_{x \rightarrow 0^+} x^\alpha \left(\frac{1}{\log\left(2x + 1 + \frac{x^3}{2}\right)} - \frac{1}{\log(1+2x)} \right)$$

$$= \lim_{x \rightarrow 0} x^\alpha \left(\frac{\log(1+2x) - \log\left(1 + 2x + \frac{x^3}{2}\right)}{\log(1+2x) \log\left(1 + 2x + \frac{x^3}{2}\right)} \right)$$

$$L = \lim_{x \rightarrow 0^+} x^\alpha \left(\frac{\ln(1+2x) - \ln\left(1 + 2x + \frac{x^3}{2}\right)}{4x^2 \cdot \left(1 + \frac{x^2}{4}\right)} \right) \frac{(2x)\left(2x + \frac{x^3}{2}\right)}{\ln(1+2x) \ln\left(1 + 2x + \frac{x^3}{2}\right)}$$

$$\Rightarrow L = \lim_{x \rightarrow 0^+} -x^{\alpha+1} \left(\frac{\ln\left(1 + \frac{x^3}{2(1+2x)}\right)}{4x^3 \left(1 + \frac{x^2}{4}\right)} \right) \frac{(2x)\left(2x + \frac{x^3}{2}\right)}{\ln(1+2x) \ln\left(1 + 2x + \frac{x^3}{2}\right)}$$

$$\text{for } L \text{ be finite non zero ; } \alpha = -1 \Rightarrow L = \frac{-1}{8}$$

$$\text{and If } \alpha = 2 ; L = 0$$

$$\text{and for } \alpha = -2 ; L \rightarrow \infty$$

$$\text{if } \alpha > -1 \text{ the limit is } 0$$

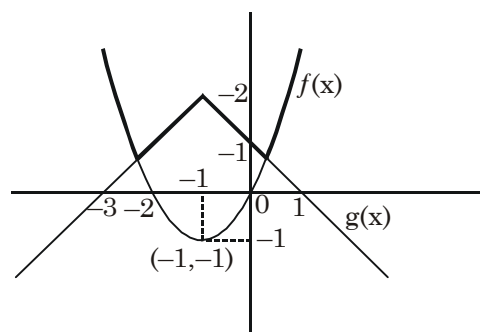
$$\text{if } \alpha = -1 \text{ the limit is } -\frac{1}{8}$$

$$\text{if } \alpha < -1 \text{ the limit is } -\infty.$$

SECTION-II

1. Ans. 5.00

Sol.



$$p = 3, q = 2$$

$$p + q = 5$$

2. Ans. 9.48 or 9.49

Sol. 1^∞ form

$$\ell = e^{\lim_{x \rightarrow 2} \left(\frac{2^x - 3^{\frac{x}{2}} - 1}{x - 2} \right)} \times \frac{4 \ln \left(1 + \frac{x}{2} - 1 \right)}{2 \left(\frac{x}{2} - 1 \right)}$$

$$\ell = e^{\lim_{x \rightarrow 2} \left(\frac{2^x - 2^2 - 3^{\frac{x}{2}} + 3}{x - 2} \right)} \times \frac{4 \ln \left(1 + \frac{x}{2} - 1 \right)}{2 \left(\frac{x}{2} - 1 \right)}$$

$$\ell = e^{(4 \ln 2 - 3 \ln \sqrt{3}) \cdot 2}$$

$$\ell = e^{2 \ln \left(\frac{16}{3\sqrt{3}} \right)} = \frac{256}{27}$$

3. Ans. 0.25

Sol. $t = y^x$

$$\ell n t = x \ell n y$$

$$\frac{dt}{dx} = t \left[\frac{x}{y} \frac{dy}{dx} + \ell n y \right]$$

$$\Rightarrow \frac{dt}{dx} = y^x \left[\ell n y + \frac{x}{y} \frac{dy}{dx} \right]$$

Similarly $s = x^y$

$$\frac{ds}{dx} = x^y \left(\frac{y}{x} + \ell n x \frac{dy}{dx} \right)$$

$$\text{So, } y^x \left(\ell n y + \frac{x}{y} \frac{dy}{dx} \right) - x^y \left(\frac{y}{x} + \ell n x \frac{dy}{dx} \right) = 0$$

when $x = 1, y = 2$

$$2 \left(\ell n 2 + \frac{1}{2} \frac{dy}{dx} \right) - 1(2 + 0) = 0$$

$$\Rightarrow 2 \ell n 2 + \frac{dy}{dx} - 2 = 0$$

$$\Rightarrow \frac{dy}{dx} - 2 = -2 \ell n 2 = \ell n \left(\frac{1}{4} \right)$$

4. Ans. 36.00

Sol. $g'(2^+) = 1 = p$

$$\text{So, } f(x) = \frac{x^3}{\sqrt{a+x}(bx - \sin x)}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{x^3}{\sqrt{a+x} \left(bx - x + \frac{x^3}{3!} - \frac{x^5}{5!} \dots \right)} = p = 1$$

$$\Rightarrow \frac{6}{\sqrt{a}} = 1 \Rightarrow a = 36; \text{ for } b = 1$$

Hence, $a.b = 36$

5. Ans. 0.25

$$\text{Sol. } 2 \sqrt{\sum_{r=1}^k r^3} = 2 \sqrt{\frac{k^2(k+1)^2}{4}} = k(k+1)$$

$$1 + 2 \sum_{r=1}^k r^3 = k^2 + k + 1$$

$$\sum_{k=1}^{\infty} \cot^{-1}(k^2 + k + 1) = \frac{\pi}{4}$$

$$3 \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \cdot \frac{\pi}{4} \right)^n = 3 \sum_{n=1}^{\infty} \left(\frac{1}{4} \right)^n = \frac{3}{1 - \frac{1}{4}} = 1$$

6. Ans. 2.50

$$\text{Sol. } \frac{1}{x+1} + \frac{2}{x^2+1} + \frac{2^2}{x^{2^2}+1} + \frac{2^3}{x^{2^3}+1} + \dots + \frac{2^n}{x^{2^n}+1}$$

$$= \frac{x-1}{x^2-1} + \frac{2}{x^2+1} + \frac{2^2}{x^{2^2}+1} + \frac{2^3}{x^{2^3}+1} + \dots + \frac{2^n}{x^{2^n}+1}$$

$$= \frac{x+1}{x^2-1} - \frac{2}{x^2-1} + \frac{2}{x^2+1} + \frac{2^2}{x^{2^2}+1} + \frac{2^3}{x^{2^3}+1} + \dots + \frac{2^n}{x^{2^n}+1}$$

$$= \frac{1}{x-1} - \left[\frac{2^2}{x^{2^2}-1} - \frac{2^2}{x^{2^2}+1} - \frac{2^3}{x^{2^3}+1} \dots - \frac{2^n}{x^{2^n}+1} \right]$$

$$S_{n+1} = \frac{1}{x-1} - \frac{2^{n+1}}{x^{2^{n+1}}-1}$$

$$\text{So for } x = 7; S_{n+1} = \frac{1}{7-1} - \frac{2^{n+1}}{7^{2^{n+1}}-1}$$

$$\text{as } \lim_{n \rightarrow \infty} S_{n+1} = \frac{1}{6} = \ell$$

$$\Rightarrow 15\ell = \frac{15}{6} = \frac{5}{2} = 2.5$$

OR

$$\text{Think } \frac{2^r}{x^{2^r}+1} = \frac{2^r(x^{2^r}+1-2)}{(x^{2^r}-1)(x^{2^r}+1)} = \frac{2^r}{x^{2^r}-1} - \frac{2^{r+1}}{x^{2^r}-1}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I)**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, B)**

$$\text{Sol. } \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{\infty} \right)$$

$$\Rightarrow f = 40 \text{ cm.}$$

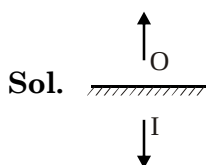
Shift due to the liquid slab

$$= t \left(1 - \frac{1}{\mu} \right) = 40 \left(1 - \frac{1}{2} \right) = 20 \text{ cm}$$

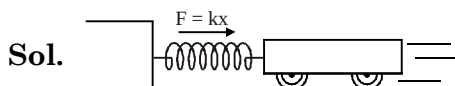
For the right hand side planoconvex lens

$$u = 40 + 20 - 40 = 20 \text{ cm}$$

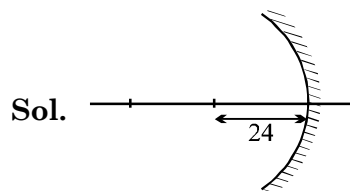
$$\frac{1}{v} - \frac{1}{20} = \frac{1}{40} \Rightarrow \frac{40}{3} v = \text{cm}$$

2. **Ans. (B,D)** V_0 is decreasing $\Rightarrow a_0$ is -ve $\Rightarrow a_I$ is in opposite direction $\Rightarrow a_I$ is +ve.3. **Ans. (A,D)**

Sol. Acceleration of block is in horizontal plane and only friction force is the force able to provide this acceleration.

4. **Ans. (A,C,D)**5. **Ans. (B,C)**

$$\vec{a} = \frac{\vec{F}}{m}$$

6. **Ans. (A,B)**

$$f = -24$$

$$|v| = 48$$

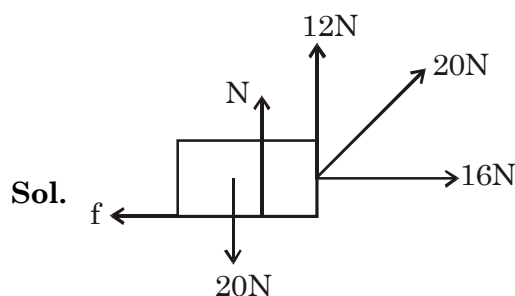
$$u = ?$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$$

$$\frac{1}{u} = \frac{1}{-24} \pm \frac{1}{48} = \frac{-2 \pm 1}{48} = -\frac{1}{48}, -\frac{3}{48}$$

$$\Rightarrow u = -48 \text{ cm or } -16 \text{ cm}$$

7. **Ans. (A,B,C,D)**

$$N + 12 = 20 \Rightarrow N = 20 - 12 = 8 \text{ N}$$

$$f = \mu N = 1.5 \times (20 - 12) = 12 \text{ N}$$

$$16 - 12 = 2a$$

$$a = 2 \text{ m/s}^2$$

$$W_F = 16 \times 2 = 32 \text{ J}$$

$$W_f = -12 \times 2 = -24 \text{ J}$$

8. Ans. (A,C,D)

9. Ans. (B)

10. Ans. (C)

11. Ans. (B)

12. Ans. (A)

SECTION-II

1. Ans. 5.00

Sol. $F_B + F_R = ma$

F_B = braking force and F_R = resistance force

$$F_B = ma - F_R$$

$$F_B = m \left(\frac{v^2}{2s} \right) - F_R$$

$$F_B = 10000 - 100 = 9900 \text{ N}$$

2. Ans. 2.00

Sol. $\frac{1}{v} - \frac{1}{25} = \frac{1}{f} \Rightarrow \frac{1}{25m_1} - \frac{1}{25} = \frac{1}{f} \dots(1)$

$$\frac{1}{v} - \frac{1}{40} = \frac{1}{f} \Rightarrow \frac{1}{40m_2} - \frac{1}{40} = \frac{1}{f} \dots(2)$$

$$m_1 = 4m_2 \quad \dots (3)$$

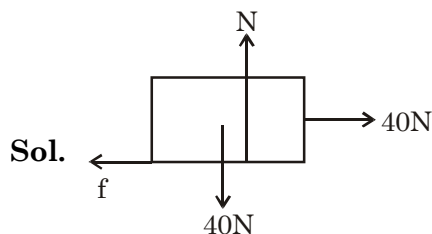
Solving $m_2 = -1$

$$\frac{1}{f} = -\frac{1}{40} - \frac{1}{40} = \frac{-1}{20}$$

$$f = -20$$

$$x = 2$$

3. Ans. 2.00



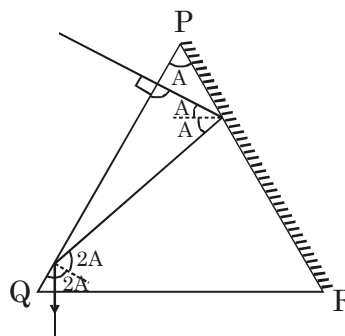
$$f_{\max} = 0.5 \times 40 = 20 \text{ N}$$

Angle of friction = θ

$$\tan \theta = \frac{20}{40} = \frac{1}{2}$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right)$$

4. **Ans. 9.00**



Sol.

$$\frac{\pi}{2} - \frac{A}{2} + \frac{\pi}{2} - 2A = \frac{\pi}{2}$$

$$\frac{5A}{2} = \frac{\pi}{2}$$

$$A = \frac{\pi}{5} = 36^\circ$$

5. Ans. 10.41 to 10.43

6. Ans. 4.00

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,B,C)
2. Ans. (A,C)
3. Ans. (B,D)
4. Ans. (A,B,C,D)
5. Ans. (A,B,D)
6. Ans. (A,B)
7. Ans. (A,C)
8. Ans. (A,C,D)
9. Ans. (C)

10. Ans. (C)
11. Ans. (A)
12. Ans. (D)

SECTION-II

1. Ans. (3.42)
2. Ans. (2.50)
3. Ans. (3.00)
4. Ans. (4.00)
5. Ans. (5.00)
6. Ans. (6.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,C,D)

Sol. $f(x) = \lim_{n \rightarrow \infty} \left(\frac{(1-x)(1+x)(1+x^2) \dots (1+x^{2^{2020}n})}{1-x} \right), |x| < 1$

$$f(x) = \frac{1}{1-x}$$

$$f\left(\frac{1}{2}\right) = 2; f\left(\frac{1}{3}\right) = \frac{3}{2}$$

$$\sin^{-1}\sin 2 = \pi - 2; \cos^{-1}\cos 2 = 2$$

2. Ans. (B,C,D)

Sol. $g'(x) = f'\left(\frac{1+2x}{1-4x}\right) \cdot \frac{(1-4x)2 - (1+2x)(-4)}{(1-4x)^2}$

$$g'(1) = f'(-1) \left(\frac{-6+12}{9} \right)$$

$$g'(1) = 3 \times \frac{6}{9} = 2$$

3. Ans. (A,C,D)

Sol. $g(x)$ is not derivable at $x = -\frac{5}{3}$ and $x = -\frac{7}{2}$.

Since $F(x)$ is derivable $\forall x \in \mathbb{R}$

$$\text{Hence } f\left(-\frac{5}{3}\right) = f\left(-\frac{7}{2}\right) = 0$$

Minimum number of roots of $f(x) = 0$ is 2.

4. Ans. (A,B,C)

Sol. $g(x) = \begin{cases} f(x) & ; x \in [0, \pi] \\ \sin x - 1 & ; x > \pi \end{cases}$

5. Ans. (C)

Sol. Let $h(x) = f^4(x) + g^4(x)$

$$h'(x) = 4f^3(x)f'(x) + 4g^3(x)g'(x)$$

$$h'(x) = 0 \Rightarrow h(x) \text{ is a constant function}$$

$$\therefore \beta = 1$$

$$\alpha = \lim_{x \rightarrow 2} \sqrt{\operatorname{sgn}\left(\frac{\{x\}}{[x]}\right)} = 1$$

6. Ans. (B,C,D)

Sol. $\lim_{x \rightarrow 0} \frac{a\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots\right) + bx\left(x - \frac{x^3}{3!} + \dots\right) + c\left(1 + x + \frac{x^2}{2!} + \dots\right) - 2x}{x^2} = 0$

$$\lim_{x \rightarrow 0} \frac{(a+c) + (c-2)x + \left(-\frac{a}{2} + b + \frac{c}{2}\right)x^2 + \dots}{x^2} = 0$$

$$\therefore a+c=0; c-2=0; -\frac{a}{2} + b + \frac{c}{2} = 0$$

$$\text{Hence } a = -2; b = -2; c = 2$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{\left(\frac{-2\cosh - 2h\sinh + 2e^h - 2h}{h^2}\right)}{h} - 0$$

$$f'(0) = \lim_{h \rightarrow 0} \left(\frac{-2\cosh - 2h\sinh + 2e^h - 2h}{h^3}\right)$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{-2\left(1 - \frac{h^2}{2!} + \frac{h^4}{4!} \dots\right) - 2h\left(h - \frac{h^3}{3!} + \dots\right) + 2\left(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots\right) - 2h}{h^3}$$

$$f'(0) = \frac{1}{3}$$

7. Ans. (B,D)

Sol. (B) By IVT it is true.

(D) By sandwich theorem it is true.

8. Ans. (B,D)

Sol. $f(\alpha) = \cos \cot^{-1} \left(\frac{\cos \alpha}{\sqrt{-\cos^2 \alpha}} \right) = \cos \cos^{-1} \left(\frac{\cos \alpha}{\sqrt{\cos^2 \alpha - \cos 2\alpha}} \right)$

Since $\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2} \right)$

$f(\alpha) = \cot \alpha$

$\frac{df(\alpha)}{d\cot \alpha} = 1$

Matching list type (Q.9 & Q.10)

9. Ans. (D)

10. Ans. (B)

Sol. $f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$

$f'(x) = 3x^2 + 2$

$f(x) = x^3 + 2x + c$

$f(0) = 1 \Rightarrow c = 1$

$f(x) = x^3 + 2x + 1$

$f'(1) = 5$

$g(x) = e^{-x} f(x)$

$g'(x) = e^{-x} f'(x) - e^{-x} f(x)$

$g'(3) = e^{-3} f'(3) - e^{-3} f(3) = -5e^{-3}$

Matching list type (Q.11 & Q.12)

11. Ans. (D)

12. Ans. (B)

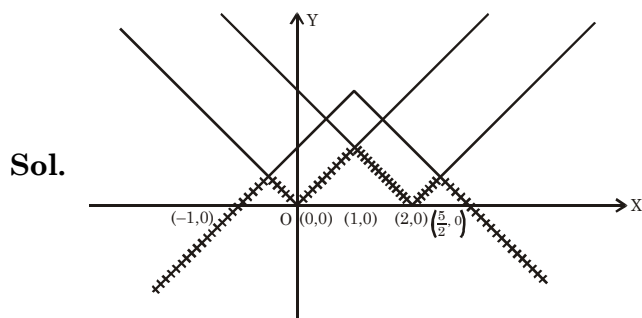
Sol. $f(x) = \begin{cases} \frac{-\sin(x-2)}{(x-2)} & ; x \in (0^-, 0) \\ e^{x-2} + 1 & ; x \in (0, 0^+) \end{cases}$

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{-\sin(x-2)}{(x-2)} = \frac{-\sin 2}{2}$

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (e^{x-2} + 1) = e^{-2} + 1$

SECTION-II

1. Ans. 5.00



$f(x)$ is non-derivable at $x = -1, 0, 1, 2, \frac{5}{2}$

2. Ans. 0.00

Sol. $|2x^2 - 3| = |3x^2 - 1|$

$2x^2 - 3 = 3x^2 - 1$ or $2x^2 - 3 = 1 - 3x^2$

$x^2 = -2$ (rejected) or $x^2 = \frac{4}{5}$ (rejected)

3. Ans. 1.80

Sol. $f(x)$ is non-derivable at $x = 0$

$[2x]$ is discontinuous at $x = \frac{1}{2}; 1; \frac{3}{2}; 2; \frac{5}{2}; 3$

$\therefore f(x)$ should be derivable & continuous at $x = 3$

$\therefore \alpha = 3; \beta = 5$ & $\gamma = 3$

$\frac{\alpha\gamma}{\beta} = \frac{9}{5} = 1.80$

4. Ans. 8.75

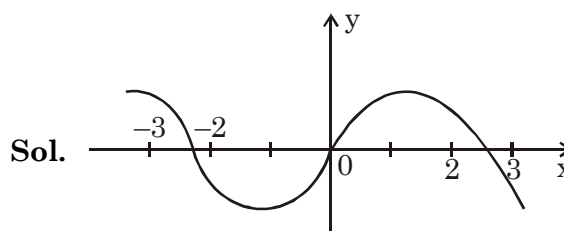
Sol. $g'(g(x)) \cdot g'(x) \big|_{x=24} = g'(g(24)) \cdot g'(24)$

$= g'(2) \cdot \frac{1}{f'(2)}$

$= \frac{1}{f'(0)} \cdot \frac{1}{f'(2)} = \frac{1}{69} = \frac{a}{b}$

$\frac{a+b}{8} = \frac{70}{8} = 8.75$

5. Ans. 3.00



$g(0) = 0$

$g(2) = 3, g(-2) = -3$

$g(3) = -4, g(-3) = 4$

$\therefore$ minimum number of roots

of $g(x) = 0$ are 3

6. Ans. 0.00

Sol. $\frac{dx}{dt} = \sqrt{\cos 2t} \cos t + \frac{-2 \sin 2t}{2\sqrt{\cos 2t}} \cdot \sin t$

$\frac{dx}{dt} \bigg|_{\pi/6} = \left(\sqrt{\cos 2t} \cos t - \frac{\sin 2t}{\sqrt{\cos 2t}} \cdot \sin t \right)$

$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2\sqrt{2}} = 0$

$\frac{dy}{dt} \bigg|_{\pi/6} \neq 0$

Hence $\frac{dx}{dy} \bigg|_{\pi/6} = 0$